

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

INARI AGRICULTURE, INC.,
Petitioner,

v.

CORTEVA AGRISCIENCE LLC,
Patent Owner.

IPR2024-01014
Patent 8,901,378 B2

Before MICHAEL J. FITZPATRICK, ZHENYU YANG, and
TIMOTHY G. MAJORS, *Administrative Patent Judges*.

Opinion for the Board filed by Administrative Patent Judge YANG.

Opinion Concurring filed by Administrative Patent Judge FITZPATRICK.

YANG, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

Inari Agriculture, Inc. (“Petitioner”) filed a Petition (Paper 2, “Pet.”) seeking *inter partes* review of claims 1–8 of U.S. Patent No. 8,901,378 B2 (Ex. 1001, “the ’378 patent”). Corteva Agriscience LLC (“Patent Owner”) filed a Preliminary Response. Paper 9 (“Prelim. Resp.”). Petitioner filed a Preliminary Reply (Paper 10), and Patent Owner filed a Preliminary Sur-reply (Paper 14, “Prelim. Sur-reply”). Trial was instituted to review the challenged claims (Paper 15, “DI”). Patent Owner filed a Request for Director Review (Paper 17), and the Acting Director denied the Request (Paper 22).

Thereafter, Patent Owner filed a Response to the Petition (Paper 24, “PO Resp.”), Petitioner filed a Reply to Patent Owner’s Response (Paper 28, “Reply”), and Patent Owner filed a Sur-reply to Petitioner’s Reply (Paper 34, “Sur-reply”). An oral hearing for this proceeding was held on August 26, 2025, and the transcript of that hearing is of record. *See* Paper 38.

The Board has jurisdiction under 35 U.S.C. § 6 and issues this Final Written Decision pursuant to 35 U.S.C. § 318 and 37 C.F.R. § 42.73. For the reasons provided below, we find Petitioner has not shown, by a preponderance of the evidence, the unpatentability of claims 1–8.

A. *Real Parties in Interest*

Petitioner identifies itself as the only real party in interest. Pet. 72. Patent Owner identifies itself and Pioneer Hi-Bred International, Inc. as the real parties in interest. Paper 4, 2.¹

¹ Paper 4 does not include any pagination. We assign the page number here as it is the second page of the Paper.

B. Related Matters

Neither party has identified any litigation related to the '378 patent. Pet. 72; Paper 4, 2; *see also* Paper 26, 1 (Patent Owner stating that it has not asserted the '378 patent against Petitioner). The parties list other patents and applications in the same family as the '378 patent. Pet. 72; Paper 4, 2.

C. The '378 Patent

The '378 patent relates to an insect resistant corn plant TC1507, the DNA plant expression construct of corn plant TC1507, and the detection of the transgene/flanking insertion region in corn plant TC1507 and progeny thereof. Ex. 1001, 1:26–31.

Before the '378 patent, insect resistance had been genetically engineered into crops such as corn to control insect damage and to reduce the need for traditional chemical pesticides. *Id.* at 1:36–38. According to the '378 patent, delta-endotoxins from *Bacillus thuringiensis* (B.t.) were successfully expressed in corn and provided excellent control over insect pests. *Id.* at 1:38–44.

The '378 patent provides a DNA construct capable of introduction into and replication in a host plant cell that, when expressed, confers insect resistance. *Id.* at 2:48–51. Specifically, the '378 patent discloses a DNA molecule named PHI8999A (SEQ ID NO:25) that includes two transgene expression cassettes. *Id.* at 2:51–3:5, 17:58–18:11. The first expression cassette comprises a promoter, 5' untranslated exon, the first intron of the maize ubiquitin (Ubi-1) gene, operably connected to a DNA molecule encoding B.t. delta-endotoxin Cry1F, and operably connected to a DNA molecule comprising a 3' ORF25 transcriptional terminator isolated from *Agrobacterium tumefaciens*. *Id.* at 2:53–62, 17:59–18:1. The second

transgene expression cassette comprises a DNA molecule of the cauliflower mosaic virus (CaMV) 35S promoter, operably connected to a DNA molecule encoding phosphinothricin acetyltransferase (PAT) gene, operably connected to a DNA molecule comprising a 3' transcriptional terminator from (CaMV) 35S. *Id.* at 2:63–3:5, 18:2–11.

The '378 patent discloses transforming maize with PHI8999A by particle bombardment and generating transgenic corn plants containing the Cry1F gene. *Id.* at 17:50–18:38. The expression of foreign genes in plants was known to be influenced by their location in the plant genome. *Id.* at 1:52–53. At the same time, the presence of the transgene at different locations in the genome will influence the overall phenotype of the plant in different ways. *Id.* at 1:57–59. For this reason, the '378 patent acknowledges that it is often necessary to screen a large number of events in order to identify an event characterized by optimal expression of an introduced gene of interest. *Id.* at 1:59–62.

According to the '378 patent,
it is common to produce hundreds to thousands of different events and screen those events for a single event that has desired transgene expression levels and patterns for commercial purposes. An event that has desired levels or patterns of transgene expression is useful for introgressing the transgene into other genetic backgrounds by sexual outcrossing using conventional breeding methods. Progeny of such crosses maintain the transgene expression characteristics of the original transformant.
Id. at 2:3–11.

In an embodiment of the '378 patent, plants that survived the transformation with PHI8999A and confirmed to express the Cry1F protein were

subjected to a whole plant bioassay using European corn borer insects. Positive plants were crossed with inbred lines to obtain seed from the initial transformed plants. A number of lines were evaluated in the field. The TC1507 event was selected from a population of independent transgenic events based on a superior combination of characteristics, including insect resistance and agronomic performance.

Id. at 18:30–45.

The '378 patent describes the sequences of the transgene insert and flanking regions of TC1507 event. According to the '378 patent, SEQ ID NO: 24 “represents the transgenic insert in maize line TC1507 as well as the sequence flanking the insertion sites,” SEQ ID NO: 25 “represents insert PHI8999A,” which was “used to transform maize line TC1507,” SEQ ID NO: 26 “represents part of the PHI8999A insert as well as flanking sequence 5' to the insert,” and SEQ ID NO: 27 “represents part of the PHI8999A insert as well as flanking sequence 3' to the insert.” *Id.* at 49–50, 59–60, 65–66, 67–68, Fig. 1.

D. Illustrative Claim

Independent claim 1 is illustrative of the claimed subject matter and is reproduced below.

1. A corn plant comprising a DNA construct, said DNA construct comprising a first and a second expression cassette, wherein said first expression cassette in operable linkage comprises
 - (a) a maize ubiquitin promoter;
 - (b) a 5' untranslated exon of a maize ubiquitin gene;
 - (c) a maize ubiquitin intron;
 - (d) a Cry1F encoding DNA molecule; and
 - (e) a 3' ORF25 transcriptional terminator; andsaid second expression cassette comprising in operable linkage

- (i) a CaMV 35S promoter;
 - (ii) a pat encoding DNA molecule; and
 - (iii) a 3' transcriptional terminator from (CaMV) 35S comprising the nucleic acid sequence of positions 5843 to 6032 of SEQ ID NO: 25; and
- wherein the first and second expression cassettes are flanked by SEQ ID NO: 26 at the 5' end and SEQ ID NO: 27 at the 3' end.

Ex. 1001, 85:24–41.

E. Asserted Challenge to Patentability

Petitioner asserts the following challenge to patentability:

Claims Challenged	35 U.S.C. § ²	Reference
1, 2, 8	102	Barbour ³
3, 4, 7	102/103(a)	Barbour
5, 6	103(a)	Barbour

As support for its patentability challenge, Petitioner relies on the Declaration of Aron Silverstone, Ph.D. Ex. 1007.

II. ANALYSIS

A. Principles of Law

To prevail in this *inter partes* review, Petitioner “shall have the burden of proving a proposition of unpatentability by a preponderance of the evidence.” 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d).

² The Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112-29, 125 Stat. 284, 287–88 (2011), amended 35 U.S.C. §§ 102 and 103, effective March 16, 2013. AIA §§ 3(b), 3(c), and 3(n). The application that issued as the ’378 patent was filed before the effective date of AIA. Ex. 1001, code 22. Accordingly, we apply the pre-AIA version of §§ 102 and 103.

³ Barbour et al., US 2006/0037095 A9, published Feb. 16, 2006 (Ex. 1009, “Barbour”).

Petitioner also bears the initial burden of production to show the asserted reference is prior art. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1379 (Fed. Cir. 2015). Once Petitioner meets that burden, Patent Owner bears the “the burden of going forward with evidence either that the prior art does not actually anticipate” or render obvious the challenged claims, or that the asserted reference is not prior art because the challenged claims are “entitled to the benefit of a filing date prior to the alleged prior art.” *Id.* Once Patent Owner presents its evidence and argument in support of the earlier filing date, the burden of going forward again shifts to Petitioner to persuade us that the challenged claims are not entitled to the benefit of the earlier filing date. *Id.* Ultimately, Petitioner bears the burden of persuasion to prove unpatentability by a preponderance of the evidence, and that burden never shifts to Patent Owner. *Id.* at 1378.

“[A] patent’s claims are not entitled to an earlier priority date merely because the patentee claims priority.” *In re NTP, Inc.*, 654 F.3d 1268, 1276 (Fed. Cir. 2011). “Rather, for a patent’s claims to be entitled to an earlier priority date, the patentee must demonstrate that the claims meet the requirements of 35 U.S.C. § 120.” *Id.*; *see also Nat. Alternatives Int’l, Inc. v. Iancu*, 904 F.3d 1375, 1380 (Fed. Cir. 2018) (holding that “claims in a patent or patent application are not entitled to priority under § 120 at least until the patent owner *proves* entitlement to the PTO, the Board, or a federal court”). Under § 120, entitlement to the benefit of an earlier priority date requires that the priority application satisfies the requirements under § 112, which include, among others, the enablement requirement. *See* 35 U.S.C. §§ 112, 120.

“The test of enablement is whether one reasonably skilled in the art could make or use the invention from the disclosures in the patent coupled with information known in the art without undue experimentation.” *United States v. Telectronics, Inc.*, 857 F.2d 778, 785 (Fed. Cir. 1988). Whether undue experimentation is needed is not a single, simple factual determination, but rather is a conclusion reached by weighing many factual considerations. *In re Wands*, 858 F.2d 731, 737 (Fed. Cir. 1988). These factors include, for example, the nature of the invention, the state of the prior art, the level of one of ordinary skill, the level of predictability in the art, and the amount of direction provided by the inventor. *Id.*

“Where an invention depends on use of living materials such as microorganisms or cultured cells, it may be impossible to enable the public to make the invention solely by means of a written disclosure.”

Wands, 858 F.2d at 735; *see also In re Lundak*, 773 F.2d 1216, 1220–21 (Fed. Cir. 1985) (“When an invention relates to a new biological material, the material may not be reproducible even when detailed procedures and a complete taxonomic description are included in the specification.”).

“The deposit of biological organisms for public availability satisfies the enablement requirement for materials that are not amenable to written description or that constitute unique biological materials which can not be duplicated.” *Ajinomoto Co. v. Archer-Daniels-Midland Co.*, 228 F.3d 1338, 1345–46 (Fed. Cir. 2000).

“Although inventions involving microorganisms or other living cells often can be enabled by a deposit, a deposit is not always necessary to satisfy the enablement requirement.” *Wands*, 858 F.2d at 736 (footnotes omitted). “No deposit is necessary if the biological organisms can be

obtained from readily available sources or derived from readily available starting materials through routine screening that does not require undue experimentation.” *Id.*; *see also* 37 C.F.R. § 1.802(b) (“Biological material need not be deposited . . . if it is known and readily available to the public or can be made or isolated without undue experimentation.”).

We analyze the asserted grounds of unpatentability in accordance with these principles.

B. Claim Construction

In an *inter partes* review, we construe a claim term “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b).” 37 C.F.R. § 42.100(b) (2020). Under that standard, the words of a claim “are generally given their ordinary and customary meaning,” which is “the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention, i.e., as of the effective filing date of the patent application.” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Petitioner argues that a person of ordinary skill in the art (“POSA”) would understand that the term “‘flanked by’ permits overlap between the expression cassette and the DNA sequence it is flanked by.” Pet. 24–26. Patent Owner does not propose any claim construction. PO Resp. 6–7.

On this record and for purposes of this Decision, we see no need to expressly construe any claim term.

C. Level of Ordinary Skill in the Art

Petitioner asserts that:

POSAs as of 2004 would have had a Ph.D. in biochemistry, cell biology, genetics, molecular biology, or a related field, and 3-5 years of experience in the design or development of transgenic

plants including genomics, proteomics, biochemistry, plant transformation and crop science, or the equivalent. Additional graduate education could substitute for professional experience, or significant experience in the field could substitute for formal education.

Pet. 11.

Patent Owner does not dispute this proposed definition of the ordinarily skilled artisan. PO Resp. 6. For purposes of this Decision, we adopt Petitioner's undisputed definition of the person of ordinary skill in the art.

D. Analysis

The '378 patent application issued from application No. 12/914,271 ("the '271 application"), filed on October 28, 2010. Ex. 1001, code 22. Barbour was published in 2006 (Ex. 1009, code 48), before the 2010 filing date of the '271 application (Ex. 1001, code 22).

The '271 application, however, claims priority, through several earlier applications, to application No. 10/837,105 ("the 2004 application"), filed April 30, 2004. Ex. 1001, code 60.⁴ It is undisputed that the 2004 application and the '271 application share the same or substantially the same specification. It is also undisputed that Barbour, the sole alleged prior art, is the 2006 publication of the 2004 application. Ex. 1009, code 21.

Petitioner contends that the challenged claims are not entitled to the priority date of the 2004 application because "the April 2004 Application fails to enable POSAs to practice the Challenged Claims without undue experimentation." Pet. 27–41. According to Petitioner, "the Challenged

⁴ The '271 application also claims priority to a 2003 provisional application. Ex. 1001, code 60. That application is not at issue in this proceeding.

Claims concern a biological material that is not reproducible even when detailed procedures and a complete taxonomic description are included in the specification.” *Id.* at 31 (quotation marks omitted). Petitioner argues that “the Challenged Claims exemplify the type of invention requiring a deposit to satisfy the statutory enablement requirement under §112.” *Id.* at 30–31. Yet, Petitioner points out that “the April 2004 Application fails to identify any deposit of biological material.” *Id.* at 32. Petitioner asserts that analysis of the *Wands* factors confirms the 2004 application’s failure to meet the enablement requirement. *Id.* at 34–36.

Petitioner argues that although the 2004 application does not satisfy the enablement requirement under § 112 as of the 2004 application’s filing date, the same application, published as Barbour, is enabling prior art under § 102 as of Barbour’s 2006 publication date. Pet. 47–57; *see also id.* at 57–60 (arguing that “the standard for enablement under §102 differs from that under §112, such that TC1507’s commercial availability is dispositive under §102 but insufficient under §112”) (heading normalized).

In the DI, the majority explained that (1) the disclosure of the 2004 application, by itself, did not sufficiently enable the full scope of the challenged claims as of April 2004; (2) because Patent Owner did not make a deposit of the transgenic TC1507 germplasm, the enablement requirement is not met by deposit of living materials. DI 10–13. During trial, Patent Owner does not dispute these determinations. *See generally* PO Resp.

Relying on the deposition testimony of Dr. Silverstone (Ex. 2003) and other evidence developed through trial, however, Patent Owner argues that the TC1507 event was “known and readily available” by April 2004, and certainly when the ’378 patent issued in 2014. PO Resp. 17–29. Thus,

according to Patent Owner, the challenged claims are entitled to the priority date of the 2004 application, which predates Barbour’s 2006 publication date. *See id.* at 1–2, 39–43. As a result, Patent Owner asserts that Barbour is not prior art and all of Petitioner’s challenges fail. *Id.* Based on the record developed during trial, and for reasons explained below, we agree with Patent Owner.

During the preliminary stage of this proceeding, Patent Owner argued that, “[f]or claims where a biological sample may be required to enable the invention, enablement can be shown as of the filing date by depositing the material or otherwise establishing it is known and readily available *as of the date of issuance* of the patent.” Prelim. Sur-reply 1 (citing *Lundak*, 773 F.2d at 1220–22; *Feldman v. Aunstrup*, 517 F.2d 1351, 1355 (CCPA 1975); *Ajinomoto Co. v. Archer-Daniels-Midland Co.*, 228 F.3d 1338, 1345–46 (Fed. Cir. 2000); 37 C.F.R. § 1.802(b); M.P.E.P. §§ 2402, 2404.01); *see also* Prelim. Resp. 18 (citing *Lundak*, 773 F.2d at 1222; 37 C.F.R. § 1.804(a)), 26–27 (citing *Ajinomoto*, 228 F.3d at 1346; *Lundak*, 773 F.2d at 1220–22; *Feldman*, 517 F.2d at 1355). Patent Owner also argued that “Petitioner’s own evidence of enablement of Barbour in 2006 for the purposes of § 102 establishes enablement *as of the April 2004 filing date*.” Prelim. Sur-reply 4.

In the DI, the majority stated that “for the priority determination, the relevant time is April 30, 2004, the filing date of the 2004 application.” DI 21; *see also* DI Dissent 3 (agreeing “Petitioner correctly argues that . . . whether the 2004 application enables the challenged claims must be determined as of April 30, 2004”). The majority explained that “[t]he authorities Patent Owner relie[d] on are inapposite because they address the deposit, and not commercial availability, of biological material.” DI 16; *see*

also id. at 16–17 (analyzing authorities cited by Patent Owner). Because “Patent Owner d[id] not point to any authority to show, or otherwise persuasively argue, that case law and Rules addressing the deposit of biological material would apply to commercial sales of the material as presented on the record here,” “[w]e decline[d] to hold that the commercial availability of biological material during the pendency of the application but after the application’s filing date (for which the priority benefit is sought) retroactively meets the enablement requirement as of the filing date.” *Id.* at 18. Based on the record available at that time, and for purposes of institution, the majority determined that Petitioner had shown sufficiently that the 2004 application did not meet the enablement requirement as of 2004, and thus, the challenged claims are not entitled to the 2004 priority date. *Id.* at 18–19.

After institution, Patent Owner continues to argue that “[i]t is sufficient to enable the claimed invention *as of the filing date* by depositing a seed or otherwise showing that the biological material is known and readily available *by the issuance of the patent*.” PO Resp. 16. In addition to the authorities cited during the preliminary stage of this proceeding, which the majority addressed and dismissed in the DI (*see* DI 16–17 (analyzing authorities cited by Patent Owner in the Preliminary Response and Preliminary Sur-reply)), Patent Owner now also calls our attention to a 1987 Federal Register publication, setting forth the Office’s position related to the deposit requirement. PO Resp. 16–18 (citing 52 Fed. Reg. 34080, 34081 (Sept. 9, 1987)).

In the Federal Register, the Office stated that “[i]f a biological material is *known and readily available at the time the patent is granted*, and

there is no reasonable basis to believe that the biological material will cease to be available during the life of the patent, no requirement for a separate deposit would be made.” 52 Fed. Reg. 34080, 34081 (emphasis added). This statement supports Patent Owner’s argument that, “[f]or claims where a biological sample may be required to enable the invention, enablement can be shown as of the filing date by depositing the material or otherwise establishing it is ‘known and readily available’ *as of the date of issuance* of the patent.” PO Resp. 17–18.

Petitioner does not challenge the authority of this Federal Register publication. In fact, Petitioner relies on other passages from the same publication. *See* Reply 3, 5, 18, 20 (citing different excerpts from 52 Fed. Reg. 34080). Petitioner, however, does not address the guidance in the Federal Register identified by Patent Owner. Instead, Petitioner argues that “[t]he Board already rejected PO’s theory and explained that PO’s authorities were ‘inapposite’ and did not concern ‘commercial availability.’” Reply 26 (citing DI 16). Petitioner, however, fails to recognize that our decision to institute was based on the record at that time. The parties have developed the record since then.

Indeed, Patent Owner has cited the Federal Register, which persuasively shows that the Office only requires “a biological material is known and readily available at the time the patent is granted” to satisfy the enablement requirement. PO Resp. 16–18 (citing 52 Fed. Reg. 34080, 34081). The record thus contradicts Petitioner’s argument that “PO still cites ‘no authority that directs [the Board] to extend the law applicable to the deposit of biological material to commercial sales of such material.’” Reply 26 (citing DI 18).

Petitioner contends that “[c]ontrolling precedent instead precludes such extension.” *Id.* (citing *White Consol. Indus. v. Vega Servo-Control*, 713 F.2d 788, 791 (Fed. Cir. 1983)). Although Petitioner is correct that *White Consolidated Industries* stands for the general rule that enablement is evaluated as of the application filing date (713 F.2d at 791), the patent-at-issue in that case “is directed to a numerical control (NC) system for machine tools,” and not a biological material. *White Consol. Indus.*, 713 F.2d at 789.

For a claim to a biological material,

the enablement requirement of § 112, first paragraph, does not require such assured access to a microorganism deposit as of the filing date; what is required is assurance of access (to the microorganism culture by the public upon issuance of a patent on the application) prior to or during the pendency of the application, so that, upon issuance of a U. S. patent on the application, the public will, in fact, receive something in return for the patent grant.

Feldman, 517 F.2d at 1355. Patent Owner has persuaded us that such exception not only applies to a deposit, but also extends to a biological material known and readily available. *See* 52 Fed. Reg. 34080, 34081.

In this case, it is undisputed that TC1507 germplasm was known and readily available to the public by Barbour’s publication date in 2006 (Pet. 49–53), years before the 2014 issuance date of the ’378 patent (Ex. 1001, code 45). Thus, based on the record developed through trial, we agree with Patent Owner that the 2004 application satisfies the enablement requirement as of its application date in 2004. PO Resp. 16–18 (citing 52 Fed. Reg. 34080, 34081). As a result, the challenged claims are entitled to the 2004 priority date and, thus, Barbour does not qualify as prior art.

Because all of Petitioner’s challenges rely on Barbour (Pet. 6), all those challenges fail.

III. CONCLUSION

After reviewing the entire record, we determine that Petitioner has not met its burden to show, by a preponderance of the evidence, that claims 1–8 of the ’378 patent are unpatentable.

In summary:

Claims	35 U.S.C. §	References	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1, 2, 8	102	Barbour		1, 2, 8
3, 4, 7	102/103(a)	Barbour		3, 4, 7
5, 6	103(a)	Barbour		5, 6

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has not shown, by a preponderance of the evidence, that claims 1–8 of the ’378 patent are unpatentable; and

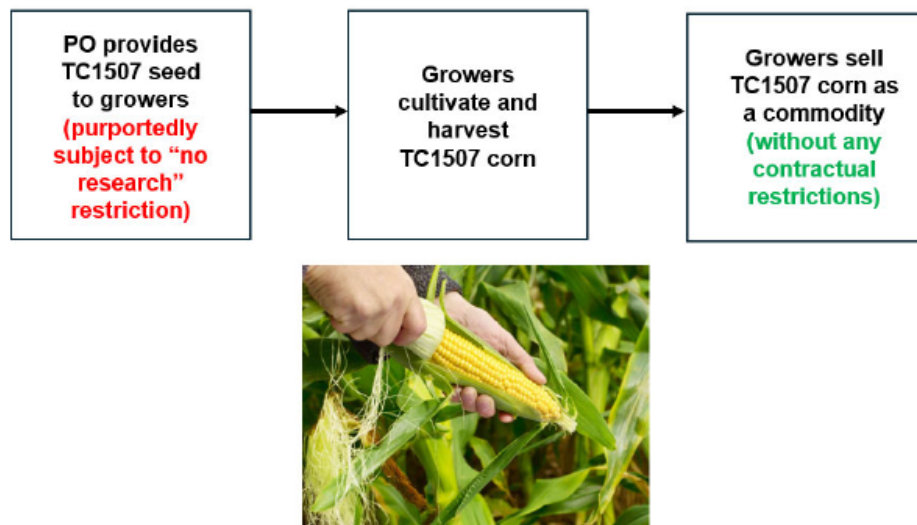
FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

FITZPATRICK, *Administrative Patent Judge*, concurring.

I join the majority opinion, but I would also hold that Petitioner has not met its burden to show, by a preponderance of the evidence, that claims 1–8 of the '378 patent are unpatentable for an independent reason.

Petitioner asserts the following in its Petition:

[C]orn harvested from TC1507 plants continued to express the TC1507 event and was indisputably commercially available following the 2003, 2004, and 2005 growing seasons. Silverstone [Ex. 1007], ¶¶134–134. Such corn was sold as a commodity—without any contractual restrictions:



Buyers of corn from growers lack any contractual relationship with PO and are thus not bound by any purported restrictions.

Pet. 52–53. Petitioner relies on these asserted facts to argue that Barbour is enabling for purposes of anticipation of some of the challenged claims.

Pet. 47. In Petitioner's words:

Barbour enables Challenged Claims 1–2 and 8 for anticipation purposes because PO had commercialized TC1507

more than a year before July 6, 2007^[1] (infra §VI.A.2.a) and POSAs could thus screen commercially available corn to identify germplasm expressing the TC1507 event, thereby possessing the claimed invention (infra §VI.A.2.b). Barbour itself discloses such screening methods and related kits.

Pet. 47.

Patent Owner, however, argues that these same facts likewise demonstrate enablement of the challenged claims by the 2004 application as of its filing date of April 30, 2004, which would eliminate Barber as prior art. *See* PO Resp. 19–20 (“Petitioner’s own evidence, presented for the purposes of institution, also demonstrates that the TC1507 event was undisputably ‘known and readily available’ by the April 2004 filing date.”).

Although the Petition does not provide a precise date within 2003 that TC1507 became publicly available, the Petition implies that it was by autumn of 2003 and certainly before the end of 2003 and thus prior to April 30, 2004. *See* Pet. 52 (“following the 2003 . . . growing season[.]”), 5 (“PO commercialized TC1507 starting in 2003—making TC1507 germplasm accessible years before the July 2006 critical date.”). The testimony of Petitioner’s declarant, Dr. Silverstone, further supports that TC1507 was known and readily available in 2003. First, Dr. Silverstone testifies that “the TC1507 event had been commercialized under Patent Owner’s ‘Herculex’ trademark and widely planted in the United States over the course of three growing seasons (i.e., 2003, 2004, and 2005).” Ex. 1007 ¶131. Second, Dr. Silverstone testified that corn harvested from these growing seasons,

¹ The significance of July, 6, 2007, is that it is the filing date of the next application in the priority chain after the 2004 application. Ex. 1001, code (60), 1:7–13.

including the 2003 growing season, was commercially available to third parties whose use of the corn was unrestricted. Specifically, Dr. Silverstone testified:

I understand that in European patent office proceedings, Patent Owner cited contractual provisions purportedly barring farmers from using TC1507 seed for “research purposes” or “transmitting the seed onwards to third parties” for such purposes. EX1003, 543 (emphasis original). However, I have been informed that such alleged restrictions would not apply to bulk corn harvested (and then sold into commerce, making it commercially available) from the TC1507 seed that farmers planted. Such corn kernels would continue to express the TC1507 event. *See* EX1003, 294 (noting that “[a]ny given event can be transmitted by breeding from the original transformant plant, thereby to create descendent plants with the same event.”).

Ex. 1007 ¶134.

I am aware of Petitioner’s arguments, in its Reply, that the amount of corn expressing the TC1507 event that was harvested and sold as a commodity following the 2003 growing season would have been less than that of the 2005 growing season and that a POSA “would [have had to] purchase such commodity corn, test it for TC1507, and hope for positive results,” which Petitioner characterizes as a “needle-in-haystack” scenario. Reply 10. And, I am persuaded that the amount of such corn harvested and sold would have been less in 2003 than in 2005. However, Petitioner offers no basis on which to quantify such corn in 2003 versus 2005, let alone explain how the 2003 amount is insufficient, but the 2005 amount is sufficient, to support enablement of the challenged claims based on the same paper disclosure.

In sum, Petitioner argues, and the record supports, that “corn harvested from TC1507 plants continued to express the TC1507 event and

was indisputably commercially available following the 2003, 2004, and 2005 growing seasons” and “[s]uch corn was sold as a commodity—without any contractual restrictions. ” Pet. 52–53. Further, Petitioner argues, and the record supports, that “POSAs could thus screen commercially available corn to identify germplasm expressing the TC1507 event, thereby possessing the claimed invention,” with “Barbour itself disclos[ing] such screening methods and related kits.” *Id.* at 47. Because of these facts, I determine that the 2004 application provides an enabling disclosure of the challenged claims and that Barber, which was published in 2006, is not prior art as to the challenged claims.

For this independent reason, I would hold that Petitioner has not met its burden to show, by a preponderance of the evidence, that claims 1–8 of the ’378 patent are anticipated by, or would have been obvious over, Barber.

IPR2024-01014
Patent 8,901,378 B2

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